

Coupled Forward ECG on Real-Geometry FEM: ASM vs. sASM Domain-Decomposition Preconditioning

Branch `cardiac-sim-fakegeo` · algorithm study (numerical linear algebra / domain decomposition)

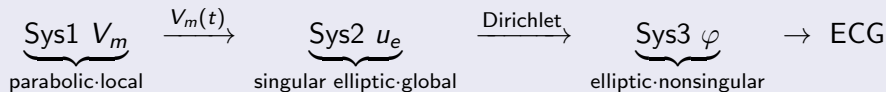
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MFEM 4.9 + PETSc 3.19 · all numbers measured on `heart.msh`

Overview: nature of the study and the main thread

Nature: a Krylov-preconditioning / domain-decomposition (Additive Schwarz) *algorithm* study. The vehicle is a coupled-PDE forward problem, but the *object of study* is *how the linear systems converge faster and scale*.

Pipeline



This deck focuses on: the problem definition, the mesh, and the **ASM(BASIC)** vs. **sASM(scaled/restricted)** **iteration counts** and **wall-clock time** on the three systems.

Problem: algebraic characterization of the three systems

Each time step solves three SPD (semi-)definite systems (`heart.msh`, Sys2 has 10085 unknowns):

System	Equation	Algebraic type	baseline iters
Sys1 V_m	$(\frac{1}{\Delta t}M + \frac{1}{2}K) V_m = b$	parabolic / mass-dominated / nonsingular	CG+ICC ~ 13
Sys2 u_e	$K_{\sigma_i + \sigma_e} u_e = -K_{\sigma_i} V_m$	elliptic / pure-Neumann singular / global	$\sim 82-96$
Sys3 φ	$K_{\sigma_o} \varphi = 0$, interface Dirichlet	elliptic / nonsingular / global	$\sim 62-66$

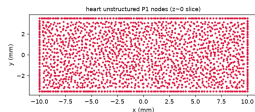
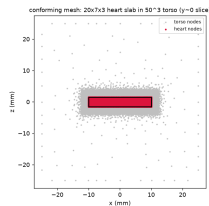
Sys2 is the only hard one (singular + elliptic + global):

- $K\mathbf{1} = 0$: null space = constants = one slow global mode $\Rightarrow$ de-mean the RHS + `MatSetNullSpace` (project the constant mode out every CG iteration).
- global elliptic $\Rightarrow$ a one-level domain decomposition propagates only one subdomain hop per iteration; more subdomains = slower (the core theme below).

Mesh: conforming unstructured tetrahedra (heart-in-torso)

Geometry: a heart slab $20 \times 7 \times 3$ mm centered inside a 50^3 mm torso; the interface is **conforming** (Gmsh BooleanFragments, sharing the same interface nodes).

Discretization: continuous P1 Galerkin (MFEM 4.9), anisotropic σ (fibers $\parallel x$), units mm/ms/mV/mS $\cdot$ mm $^{-1}$.



mesh	nodes (dof)	tets	bdr tris
heart	10 085	47 582	9 510
torso	36 229	206 647	12 112

interface shared dofs: 4 757 (coord-matched, partition-independent)

Subdomains = MPI ranks (METIS geometric partition); $n_p = \# \text{ranks} = \# \text{subdomains}$.

Method: ASM(BASIC) vs. sASM(scaled / restricted)

One-level additive Schwarz, blocks = METIS subdomains, sub-solve = ICC(0):

$$M_{\text{ASM}}^{-1} = \sum_i R_i^{\top} A_i^{-1} R_i, \quad A_i = R_i A R_i^{\top} \text{ (overlap } O \text{ layers)}$$

The key difference — overlap *multiplicity* (over-count)

- **ASM(BASIC)**: overlap dofs get *one contribution from each* overlapping subdomain $\Rightarrow$ **over-counting** in the overlap, which corrupts the operator spectrum and can even *increase iterations with overlap*.
- **sASM**: a diagonal partition of unity $D^{-1/2}(\cdot)D^{-1/2}$ cancels the over-count ($D =$ multiplicity) $\Rightarrow$ overlap works *as intended*, iterations drop with overlap.

$$M_{\text{sASM}}^{-1} = D^{-1/2} \left(\sum_i R_i^{\top} A_i^{-1} R_i \right) D^{-1/2}$$

At $O=0$ (no overlap) there is no over-count, so **ASM** $\equiv$ **sASM**.

Iterations: ASM vs. sASM ($n_p=4$, $\text{rtol } 10^{-8}$, ICC(0))

System	O	ASM(BASIC)	sASM	note
Sys1 V_m	0	11	11	$O=0$: equal
	1	13 $\uparrow$	8 $\downarrow$	
	2	13 $\uparrow$	8 $\downarrow$	
Sys2 u_e (singular)	0	82	82	
	1	92 $\uparrow$	74 $\downarrow$	
	2	93 $\uparrow$	74 $\downarrow$	
Sys3 φ	0	62	62	
	1	88 $\uparrow$	53 $\downarrow$	
	2	94 $\uparrow$	53 $\downarrow$	

$\uparrow$ ASM iterations *rise with overlap* (over-count); $\downarrow$ sASM *falls* with overlap (overlap works correctly).

Iterations: ASM vs. sASM ($n_p=8$, more subdomains $\Rightarrow$ stronger anomaly)

System	O	ASM(BASIC)	sASM	note
Sys1 V_m	0	13	13	
	1	18 $\uparrow$	8 $\downarrow$	
	2	19 $\uparrow$	8 $\downarrow$	
Sys2 u_e (singular)	0	79	79	
	1	117 $\uparrow$	73 $\downarrow$	
	2	123 $\uparrow$	71 $\downarrow$	
Sys3 φ	0	66	66	
	1	93 $\uparrow$	55 $\downarrow$	
	2	109 $\uparrow$	53 $\downarrow$	

Most extreme: Sys3 ASM $O:0 \rightarrow 2$ **66 $\rightarrow$ 93 $\rightarrow$ 109 (+65%)**; sASM **66 $\rightarrow$ 55 $\rightarrow$ 53 (−20%)**.

Wall-clock: ASM vs. sASM (ms; KSP+PCASM setup + solve, avg of 20)

$n_p = 4$

System	O	ASM	sASM
Sys1	0	3.66	3.79
	1	5.27	5.60
	2	6.88	5.22 ↓
Sys2	0	15.39	18.85
	1	21.50	17.60 ↓
	2	24.33	19.39 ↓
Sys3	0	43.59	46.99
	1	68.10	44.36 ↓
	2	80.18	52.57 ↓

$n_p = 8$

System	O	ASM	sASM
Sys1	0	5.37	5.62
	1	8.93	7.89 ↓
	2	14.90	8.91 ↓
Sys2	0	24.48	26.73
	1	45.72	28.94 ↓
	2	53.15	32.83 ↓
Sys3	0	72.98	59.58↓
	1	96.81	71.03 ↓
	2	133.24	77.17 ↓

For $O \geq 1$, sASM is **both fewer iterations and faster wall-clock**; for ASM, overlap *costs time and raises iterations*. $n_p=8$ Sys3 $O=2$: ASM 109it/133ms vs. sASM 53it/77ms ($\sim 1.7\times$).

Key finding: the overlap anomaly

Observation

Adding overlap to one-level **ASM(BASIC)** *raises* iterations and slows wall-clock; adding overlap to **sASM** *lowers both*.

Mechanism: overlap anomaly = over-count $\times$ inexactness

- overlap dofs counted once per subdomain $\Rightarrow$ BASIC **amplifies** in the overlap ($\|M^{-1}\|$ too large), degrading the spectrum.
- ICC(0) sub-solves are *inexact*, so the amplified error is not absorbed by an exact local solve $\Rightarrow$ iterations climb.
- the $D^{-1/2}$ partition of unity in sASM cancels the over-count exactly $\Rightarrow$ restores the correct behavior “more overlap = faster information transport”.

Conclusion: on real-geometry FEM, **scaled/restricted ASM is the correct way to use overlapping DD**; BASIC overlap is harmful here. More subdomains ($n_p:4 \rightarrow 8$) $\Rightarrow$ stronger anomaly, underscoring that *scalability* needs the correct weighting (and, later, a coarse space).

Summary and outlook

- **Problem:** coupled forward ECG on three systems; Sys2 (singular pure-Neumann elliptic) is the only hard one and the target of the preconditioning study.
- **Mesh:** conforming unstructured tet P1-FEM, heart 10085 dof / 47582 tets, torso 36229 / 206647.
- **ASM vs. sASM** (measured, $n_p=4, 8$):
 - ASM(BASIC) iterations **rise with overlap** (over-count), wall-clock slower;
 - sASM iterations **fall with overlap**, and wall-clock is faster for $O \geq 1$;
 - at $O=0$ the two coincide.
- **Next steps** (implemented / planned on the same branch): strong fine level **SORAS** ($84 \rightarrow 20$, $4.2\times$), shared **Nicolaides coarse space** (scalability), cross-time **Fischer** (-12%); at scale (~ 3000 cores) a **GenEO spectral coarse space** + a telescoped scalable coarse solve.

All numbers measured in-container (MFEM 4.9 + PETSc 3.19); ECG bit-identical (verified).